ATMIYA DEVELOPER STUDENTS CLUB PRESENTS

CODE CARNIVAL 3.0

QualityIQ: Adaptive Industrial Quality Intelligence Platform

"Don't just detect defects. Decide quality, recover from bad inspections, and explain what is happening on the line."

TEAM NAME	TEAM ID	TEAM LEADER
[Your team name]	[ID from the ADSC portal]	[Full name]

PROBLEM STATEMENT TITLE	THEME / TRACK
Industrial Visual Quality Inspection with AI	AI and Computer Vision / Smart Manufacturing

01 · THE TARGET

PROBLEM STATEMENT & UNDERSTANDING

PROBLEM STATEMENT

- Surface defects, missing parts, and orientation errors bypass manual QC
- High line speeds cause severe operator fatigue, missed flaws, and scrap waste
- Moving parts, glare, and vibration trigger false rejects in naive AI detectors
- Directly impacts manufacturing plant managers, QC inspectors, and assembly lines

OUR UNDERSTANDING

- Root cause: Factory vision needs decisions and recovery, not just raw detection
- Existing gap: Binary bounding-box models fail on blurred frames and unseen defects
- Critical zones matter: A scuff on a case is cosmetic; a bent lead shorts the board
- Why now: Edge AI inference now enables sub-50ms closed-loop decisions on lines

02 · THE PLAN

PROPOSED SOLUTION

OUR IDEA

- 3-layer architecture: Perception -> Decision Engine -> Quality Intelligence
- Classical optical Quality Gate filters blur, glare, and exposure before inference
- Automated multi-camera Recovery Loop re-inspects flawed frames before decision
- Calibrated 4-state verdict: PASS, REJECT, REVIEW, and UNINSPECTABLE

THREE LAYERS, ONE CLOSED LOOP

1. PERCEPTION LAYER

OpenCV Quality Gate (Blur/Exposure) · PatchCore Anomaly · Defect Typer

2. DECISION LAYER

Profile Rules · Safety-Critical Zones · Severity Multiplier · Recovery Loop

3. INTELLIGENCE & ACTION LAYER

CUSUM Early Warning · Contributing-Factor Lift · Retraining Candidate Pool

PASS | REJECT | REVIEW | UNINSPECT

03 · THE ARSENAL

KEY FEATURES / FUNCTIONALITY

01 Quality Gate & Recovery Loop

Pre-inference OpenCV validation checks blur, glare, and exposure. Automatically engages CAM-02/03 adaptive optical recovery before failing.

02 Four-State Decision Engine

Replaces binary pass/fail with PASS, REJECT, REVIEW, and UNINSPECTABLE. Evaluates safety-critical zones to escalate severity accurately.

03 PatchCore Multi-Defect Vision

Sub-50ms CPU inference using WideResNet-50 memory banks. Generates pixel saliency heatmaps and prototype-based defect classification.

04 Early Warning & Root-Cause Actions

Sequential CUSUM change-point tracking detects machine tool wear before scrap limits breach. Computes factor lift across machines and batches.

04 · THE EXECUTION

HOW THE SOLUTION WILL WORK

STEP 1

Capture Image

High-speed optical sensor capture on moving conveyor line

STEP 2

Quality Gate

Laplacian variance & exposure check filters bad frames

STEP 3

Vision & Decision

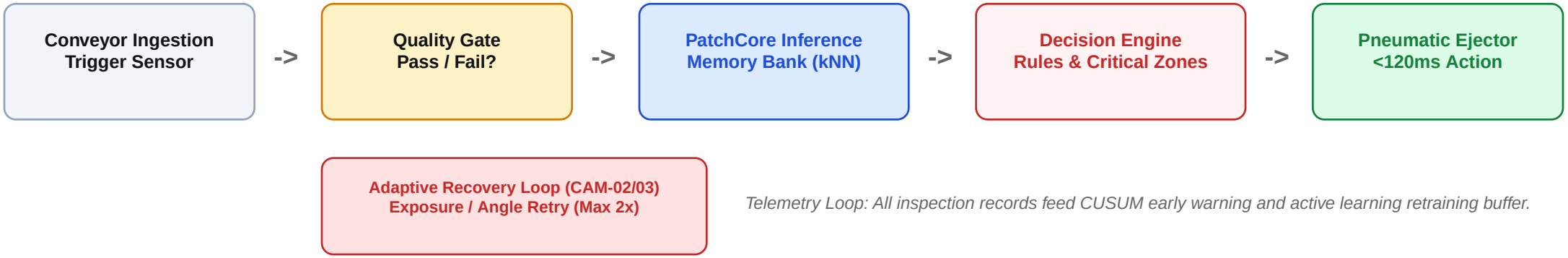
PatchCore anomaly scoring + critical-zone severity logic

STEP 4

Verdict & Action

4-state verdict triggers pneumatic pusher or QC buffer

END-TO-END PIPELINE & ADAPTIVE RECOVERY ARCHITECTURE



Telemetry Loop: All inspection records feed CUSUM early warning and active learning retraining buffer.

05 · THE GEAR

TECHNOLOGY / TECH STACK

Layer	Technology	Why we chose it
Frontend	**Next.js, TypeScript, Tailwind CSS**	High-performance enterprise dashboard with 8px rhythm and tabular figures
Backend	**Python 3.11, FastAPI, Pydantic v2**	Sub-millisecond async REST API and real-time Server-Sent Events (SSE) streaming
Database	**PostgreSQL, SQLAlchemy, JSON Store**	Immutable audit logs, serialized inspection records, and shift telemetry
AI / CV	**PyTorch, ONNX Runtime, OpenCV**	PatchCore memory banks, 40ms CPU inference, MVTec AD benchmark dataset*
Hosting	**Docker Compose, Nginx Reverse Proxy**	One-command production deployment, isolated networking, and edge gateway
Tooling	**Git, pytest, GitHub Actions, Alembic**	Deterministic regression testing, CI/CD validation, and schema migrations

**Dataset Credit: MVTec AD (MVTec Software GmbH) used under non-commercial research licence. Covers transistor and metal_nut inspection categories.*

06 · INSIDE THE MINT

CURRENT PROGRESS / WORK DONE

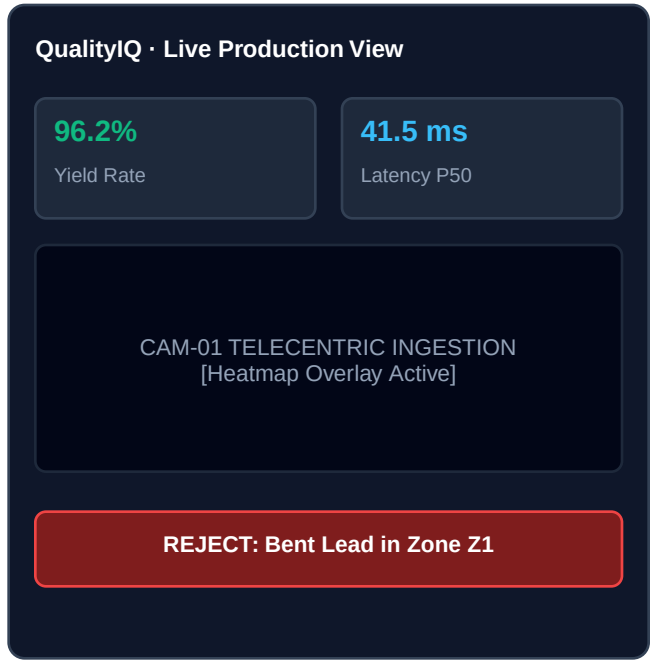
COMPLETED

- 15 industrial failure modes scoped and mapped to rules
- Complete 3-layer architecture & 4-state verdict designed
- Inspection profiles for transistor & metal_nut created
- UI/UX design specification & REST/SSE API contract defined

IN PROGRESS

- PatchCore memory bank training & tuning on MVTec AD
- Interactive 6-page dashboard & conveyor simulation build
- Unit tests for quality gate, decision logic, and CUSUM
- Docker Compose multi-container deployment packaging

PROTOTYPE PREVIEW



Design mock-up (live build in progress)

07 · THE GETAWAY

FUTURE PLAN / NEXT STEPS

PHASE 1

DURING THE HACKATHON

- Train PatchCore memory banks on transistor and metal_nut
- Deliver full interactive 6-page dashboard with fault injection
- Complete test suite and verify end-to-end Docker Compose stack

PHASE 2

RIGHT AFTER

- Pilot testing with real factory conveyor image streams
- Expand product catalog to cable, zipper, and food packaging
- Direct hardware integration with Basler GigE cameras and PLCs

PHASE 3

LONG-TERM VISION

- Multi-plant telemetry synchronization and federated learning
- Continuous active learning from operator review overrides
- SaaS and per-inspection-line licensing model for Tier-1 suppliers

08 · THE VAULT

GITHUB REPOSITORY LINK

`https://github.com/[username]/[repository]`

PUBLIC

Repository visibility is set to public with complete open-source reproduction steps.

README

Covers setup steps, 15 failure modes, system architecture, decision rules, and team.

COMMITS

Regular commits, pull requests, and verified contributions from every team member.

DEMO (OPTIONAL)

[Live link or demo video link]